

变式 6 [中档 (知识点二)] 已知 $A(-1,3), B(3,3), C(1,2\sqrt{3}+3)$, 证明 $\triangle ABC$ 是等边三角形.

6. [答案] 证明见详解.

[详解] 由两点间距离公式:

$$|AB| = \sqrt{(3+1)^2 + (3-3)^2} = \sqrt{16} = 4;$$

$$|AC| = \sqrt{(1+1)^2 + (2\sqrt{3}+3-3)^2} = \sqrt{4+12} = 4;$$

$$|BC| = \sqrt{(1-3)^2 + (2\sqrt{3}+3-3)^2} = \sqrt{4+12} = 4.$$

$|AB| = |AC| = |BC|$, 故 $\triangle ABC$ 是等边三角形.